Two Endpoint Counterexamples to the Printed Erdős-Deep Triple Classification

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Abstract

We give primitive Erdős-deep families of three modular arithmetic progressions with length triples $(10, 9, 5)$ and $(20, 5, 5)$, respectively in $\mathbb{Z}_{34}$ and $\mathbb{Z}_{48}$. Each constituent is itself Erdős-deep. Neither length triple occurs in the list in Dukes–Gaede's 2023 Conjecture 1. Both constructions use a constituent of length $\lfloor \mathrm{ord}(g)/2 \rfloor + 1$, permitted by the standing convention in Gaede's thesis, Section 2.2. They do not refute the more restrictive variant requiring every length to be at most $\lfloor \mathrm{ord}(g)/2 \rfloor$. Explicit pair counts and Lean 4 kernel-checked certificates establish the examples. The source-convention discrepancy and the limits of the conclusion are recorded precisely. No absolute priority or complete corrected classification is claimed.

1 Definitions and the endpoint

For $n \geq 1$, set

$$\mathrm{AP}_n(g, k) = \{jg \bmod n : 0 \leq j < k\}, \qquad \mathrm{dist}_n(x, y) = \min((x - y) \bmod n, (y - x) \bmod n).$$

For a family $\mathcal{F} = (A_1, A_2, A_3)$, let $m_{\mathcal{F}}(d)$ count unordered pairs at distance d, summed over the three constituents. No pairs between different constituents are counted. The family is *Erdős-deep* with parameter K if its positive distance multiplicities are precisely $1, \ldots, K - 1$, each once. It is *primitive* if $\gcd(n, g_1, g_2, g_3) = 1$.

Write $q = \mathrm{ord}(g) = n/\gcd(n, g)$. The constituent condition used here, with $k \geq 3$, is

$$k \leq \lfloor q/2 \rfloor + 1, \quad \text{equivalently} \quad 2(k - 1) \leq q. \tag{1}$$

For $1 \leq j < k$, there are exactly $k - j$ index pairs separated by j. Under (1), their distance classes $|jg|_n$ are distinct: a coincidence would give $j \equiv \pm\ell \pmod{q}$; for distinct $j, \ell \in \{1, \ldots, k-1\}$, neither $j - \ell$ nor $j + \ell$ can be a multiple of q. The points are distinct as well. Thus the constituent has multiplicities $1, \ldots, k - 1$, including at the endpoint in (1).

2 A counterexample in $\mathbb{Z}_{34}$

Theorem 1. *The family*

$$\mathcal{F}_{34} = \{\mathrm{AP}_{34}(1, 10), \mathrm{AP}_{34}(6, 9), \mathrm{AP}_{34}(12, 5)\}$$

is primitive and Erdős-deep with parameter $K = 14$. All its constituents are individually Erdős-deep.

Proof. The actual point sets, written in progression order, are

$$\begin{aligned} A &= \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}, \\ B &= \{0, 6, 12, 18, 24, 30, 2, 8, 14\}, \\ C &= \{0, 12, 24, 2, 14\}. \end{aligned}$$

Their generator orders are $34, 17, 17$. Condition (1) reads $18 \le 34$, $16 \le 17$, and $8 \le 17$, so each constituent has its asserted cardinality and is individually Erdős-deep. The first generator is 1, giving primitivity.

For B, the distances for index gaps $1, \ldots, 8$ and their weights are

$$(6, 12, 16, 10, 4, 2, 8, 14), \qquad (8, 7, 6, 5, 4, 3, 2, 1),$$

respectively. For C, the distances are $(12, 10, 2, 14)$, with weights $(4, 3, 2, 1)$. Combining these with the interval A gives every entry below.

Distance d	$m_A(d)$	$m_B(d)$	$m_C(d)$	$m_{\mathcal{F}_{34}}(d)$
1	9	0	0	9
2	8	3	2	13
3	7	0	0	7
4	6	4	0	10
5	5	0	0	5
6	4	8	0	12
7	3	0	0	3
8	2	2	0	4
9	1	0	0	1
10	0	5	3	8
12	0	7	4	11
14	0	1	1	2
16	0	6	0	6

All omitted distances have multiplicity zero. The last column is a permutation of $1, \ldots, 13$. As a separate total-count check, $\binom{10}{2} + \binom{9}{2} + \binom{5}{2} = 45 + 36 + 10 = 91 = \binom{14}{2}$. □

3 A counterexample in $\mathbb{Z}_{48}$

Theorem 2. *The family*

$$\mathcal{F}_{48} = \{\mathrm{AP}_{48}(1, 20), \mathrm{AP}_{48}(6, 5), \mathrm{AP}_{48}(18, 5)\}$$

is primitive and Erdős-deep with parameter $K = 21$. All its constituents are individually Erdős-deep.

Proof. The constituents are

$$A = \{0, 1, \ldots, 19\}, \quad B = \{0, 6, 12, 18, 24\}, \quad C = \{0, 18, 36, 6, 24\}.$$

Their generator orders are $48, 8, 8$; the three instances of (1) are $38 \le 48$, $8 \le 8$, and $8 \le 8$. They are individually Erdős-deep, and the first generator again ensures primitivity.

The interval A gives $m_A(d) = 20 - d$ for $1 \le d \le 19$, and zero otherwise. For B, the four index-gap distances are $(6, 12, 18, 24)$; for C, they are $(18, 12, 6, 24)$. Both have index-gap weights $(4, 3, 2, 1)$, so their combined contributions are

d	6	12	18	24
$m_B(d)$	4	3	2	1
$m_C(d)$	2	3	4	1
$m_B(d) + m_C(d)$	6	6	6	2

Consequently the multiplicities at distances $6, 12, 18$ change from $14, 8, 2$ to $20, 14, 8$, respectively, and distance 24 supplies the new multiplicity 2. All other multiplicities of A stay unchanged. The result is exactly $1, \ldots, 20$, each once. The pair-count check is $\binom{20}{2} + 2\binom{5}{2} = 190 + 20 = 210 = \binom{21}{2}$. □

4 Consequence and exact source scope

The fifteen triples in the printed classification [1, Conjecture 1, p. 14] do not include $(10, 9, 5)$ or $(20, 5, 5)$. Theorems 1 and 2 therefore refute its necessity claim, even after imposing primitivity, minimum constituent length three, and individual Erdős-deepness.

The endpoint issue must be separated from the exact computations. The paper's Theorem 1 [1, p. 3] includes the $+1$ endpoint in its characterization of individually Erdős-deep APs. Gaede's standing convention [2, §2.2, p. 16] allows $k_i \le \lfloor \mathrm{ord}(g_i)/2 \rfloor + 1$. Conjecture 4.5 [2, pp. 14,42] also displays the $+1$ bound for the largest constituent. An earlier general convention [2, Remark 1.10, p. 5] instead omits $+1$; applying that stricter convention to every constituent excludes both examples. The displayed search pseudocode [2, p. 43] explicitly bounds only k_1 using g_1. Both examples satisfy that displayed bound. Thus the pseudocode *does not establish why the examples were omitted*, and the implementation cause remains unknown. The all-constituent strict-half-order variant is neither proved nor disproved here.

5 Formal verification and reproducibility

The accompanying `ErdosDeepCounterexamples.lean` defines modular point lists, the cyclic metric, and a structurally recursive list of unordered-pair distances. On an input $x :: L$, that list contains the distances from x to every element of L, followed by the pair-distance list of L. Distinctness of every example's point list is separately checked, so positions correspond to actual distinct elements.

For each $d = 1, \ldots, \lfloor n/2 \rfloor$, the development counts occurrences of d. It removes zero counts and sorts the remaining multiplicities by insertion sort. Equality with the list $[1, \ldots, K-1]$ is the exact depth predicate. The following are kernel-checked:

- primitive gcd, ordering, minimum lengths, and the half-order bounds;
- distinct point lists and the complete histograms;
- individual depth of all six constituents and total depth of both families;
- exclusion from the fifteen listed triples and the negation of an explicitly normalized universal classification statement.

The checks use Lean 4.32.2, mathlib commit

`905b95818eb32af7874a58b427f50c1711a5e96c`.

There is no `sorry`, `native_decide`, or added axiom. The histogram and individual-depth theorems have no axioms; the bundled counterexamples and negated classification use only the standard `propext` and `Quot.sound` principles. The script `check-lean.sh` rebuilds the certificate, and `lean-build.log` records the axiom audit.

A C++ search through $n \leq 60$ produced the fifteen listed length triples and these two additional triples. A separate Python program enumerated unordered pairs of the actual modular points and checked all 649 recorded representative hits. This bounded search is supplemental evidence, not a classification theorem; neither counterexample depends on its exhaustiveness.

Prior-art check. A limited public-web search on 14 October 2026 used the paper's exact title, the authors and triple-classification terminology, and the two extra length triples in spaced and unspaced forms. It located the original article, thesis, and authors' research pages, but no relevant earlier report of these examples. That search is not exhaustive and does not establish priority. Its queries and checked primary URLs are recorded in the package.

References

[1] P. J. Dukes and T. Gaede, *Families of modular arithmetic progressions with an interval of distance multiplicities*, Integers **23** (2023), #A25. https://math.colgate.edu/~integers/x25/x25.pdf.

[2] T. Gaede, *Erdős-Deep Families of Arithmetic Progressions*, M.Sc. thesis, University of Victoria, 2022. https://hdl.handle.net/1828/14160.